

METLAKATLA, AK, USA

WMO: 703985

Lat: 55.131N

Lon: 131.578W

Elev: 0

StdP: 101.33

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25377

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.6	-3.0	-22.7	0.5	2.0	-17.4	0.8	1.1	11.7	4.6	10.7	4.6	4.7	80	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.3	21.1	15.5	18.9	14.9	17.6	13.8	16.2	19.8	15.3	18.0	14.7	17.2	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.1	17.0	13.8	9.8	16.6	13.0	9.3	15.8	45.0	19.8	42.7	18.3	40.8	17.1	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.0	8.0	-8.0	23.6	3.3	2.8	-10.4	25.6	-12.3	27.2	-14.2	28.8	-16.6	30.8
			-10.1	17.2	2.8	1.3	-12.1	18.1	-13.7	18.9	-15.3	19.6	-17.4	20.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	3.8	3.8	4.2	6.5	9.6	12.5	14.5	14.7	12.2	8.7	5.4	3.8
	DBStd	4.85	3.63	2.80	2.87	2.24	2.48	2.23	2.03	1.82	1.68	2.31	2.73	3.26
	HDD10.0	1079	192	175	179	109	37	2	0	0	2	52	138	194
	HDD18.3	3656	451	408	437	356	271	177	122	115	183	299	387	451
	CDD10.0	471	1	1	1	3	25	76	138	145	69	11	1	2
	CDD18.3	5	0	0	0	0	0	1	2	2	0	0	0	0
	CDH23.3	9	0	0	0	0	1	5	2	1	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	3.4	4.2	3.7	3.8	3.2	3.0	2.8	2.6	2.5	2.8	3.5	4.0	4.3
Precipitation	PrecAvg	3019	318	237	239	206	162	135	144	213	307	386	349	323
	PrecMax	3698	516	469	437	436	414	223	295	422	539	668	592	612
	PrecMin	2172	98	97	107	52	8	55	20	75	118	141	209	98
	PrecStd	397	96	97	82	95	96	51	65	101	102	126	90	123
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	12.2	12.4	16.2	20.9	22.8	22.8	22.7	18.5	15.9	12.3	16.3
		MCWB	6.7	6.4	7.7	8.5	11.8	15.4	16.8	16.3	14.1	10.5	9.5	5.7
	2%	DB	10.0	9.9	10.1	12.5	17.1	19.1	21.0	20.3	17.1	13.6	10.9	10.9
		MCWB	7.6	6.4	6.6	7.9	10.2	14.3	16.1	15.6	13.6	10.7	9.4	7.3
	5%	DB	8.8	8.0	8.7	11.1	14.1	17.4	18.8	18.7	15.9	12.5	9.8	8.9
		MCWB	7.0	5.6	6.2	7.5	9.2	13.2	15.2	15.3	12.9	10.2	8.4	6.6
	10%	DB	7.8	7.5	7.7	9.1	12.8	16.0	17.5	17.5	14.4	12.0	8.8	7.7
		MCWB	6.1	5.3	5.4	6.5	8.9	12.5	14.1	14.3	12.0	9.9	7.4	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	8.9	8.8	10.4	13.0	16.1	17.4	17.0	15.5	12.5	10.6	8.5
		MCDB	11.1	10.3	11.2	14.2	17.8	22.0	21.6	20.4	17.5	13.9	11.2	9.8
	2%	WB	8.0	7.4	7.6	8.8	11.5	14.6	16.3	16.1	14.3	11.6	9.6	7.9
		MCDB	9.0	8.1	8.8	11.4	15.1	18.5	20.1	19.4	16.3	12.6	10.7	8.9
	5%	WB	7.5	6.3	6.7	7.8	10.6	13.7	15.3	15.4	13.5	10.7	8.3	7.3
		MCDB	8.2	8.0	8.2	9.6	13.4	17.0	18.2	17.9	15.2	11.9	9.4	8.4
	10%	WB	6.6	5.4	5.7	7.1	10.0	12.7	14.6	14.8	12.7	10.0	7.6	6.3
		MCDB	7.8	7.5	7.7	8.8	12.5	15.5	17.2	17.2	14.2	11.4	8.5	8.4
Mean Daily Temperature Range	5% DB	MDBR	3.5	3.7	4.1	4.6	4.8	4.6	4.3	4.3	3.9	3.7	3.4	3.3
		MCDBR	4.5	4.6	5.7	6.9	7.3	7.8	7.2	6.9	5.4	4.5	3.9	4.7
		MCWBR	3.2	3.0	3.7	3.9	3.9	4.2	3.8	3.7	3.2	3.0	3.4	3.0
	5% WB	MCDBR	3.6	4.0	5.0	5.6	6.2	7.2	6.3	5.9	4.7	3.7	3.7	3.7
		MCWBR	3.3	3.2	3.8	3.7	3.6	4.1	3.7	3.5	3.2	3.1	3.6	3.2
Clear-Sky Solar Irradiance	taub	0.300	0.326	0.335	0.366	0.364	0.368	0.375	0.360	0.345	0.336	0.309	0.294	
	taud	2.380	2.386	2.438	2.358	2.436	2.457	2.472	2.541	2.560	2.478	2.350	2.280	
	Ebn at Noon	644	755	842	857	880	878	864	859	826	738	614	544	
	Edh at Noon	52	73	86	107	105	104	101	89	77	66	52	45	
All-Sky Solar Radiation	RadAvg	0.47	1.26	2.24	3.69	4.64	4.83	4.32	3.77	2.46	1.34	0.60	0.37	
	RadStd	0.06	0.25	0.37	0.28	0.47	0.38	0.55	0.37	0.33	0.22	0.10	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page